

Power Supply and Reference Clock Board for the AD9959/PCBZ Evaluation Platform

Design, Verification and Assembly Documentation

Chuwei Liao

Department of Physics and Astronomy, Rice University
Brockman Hall for Physics

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Abstract

This document describes a custom USB-C powered supply and reference clock board for the Analog Devices AD9959/PCBZ quad direct digital synthesis (DDS) evaluation platform. The board generates the four supply rails the evaluation board requires and provides a 125 MHz ultra-low-phase-noise reference clock, enabling operation at the AD9959's maximum 500 MHz system clock via a $4\times$ internal PLL multiplication. The design targets RF generation in the 100 to 200 MHz band for acousto-optic modulator drive in cold-atom experiments, where close-to-carrier phase noise is the governing specification. All five supply rails are linearly regulated; a passive damping network addresses USB hot-plug transients. The design has been verified by SPICE simulation across component and operating tolerance corners using component values and routed trace resistances extracted directly from the fabrication database.

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1 Scope and Purpose

The AD9959/PCBZ evaluation board requires four externally supplied voltage rails and an external reference clock. It provides neither. The standard laboratory approach—four bench supplies and a signal generator—occupies significant rack space, introduces ground loops through multiple mains-referenced instruments, and ties up a synthesizer permanently.

This board replaces that arrangement with a single USB-C connection. It generates all four rails and supplies a fixed 125 MHz reference derived from a crystal oscillator selected for phase noise rather than frequency accuracy.

1.1 Design Objectives

1. Supply all AD9959/PCBZ rails from a single USB-C source.
2. Deliver a reference clock permitting 500 MHz system clock operation, the device maximum.
3. Minimise phase noise at DDS output frequencies of 100 to 200 MHz.
4. Avoid switching regulators in any path feeding analogue supplies.
5. Survive USB hot-plug transients without exceeding regulator absolute maximum ratings.

2 Requirements

2.1 Supply Rails

Table 1 lists the requirements extracted from the AD9959 data sheet (Rev. C) and the EVAL-AD9959 user guide (Rev. A). Analog Devices does not publish a current figure for the VCC_USB rail; the value given is estimated from the evaluation board schematic.

Table 1: AD9959/PCBZ supply requirements and delivery point

Rail	Voltage	Current	Eval. board entry	Source
AVDD	1.8 V $\pm$ 5%	225 mA [†]	J10/J16–J20 (coax)	Data sheet Table 1
DVDD	1.8 V $\pm$ 5%	145 mA	TB1 pin 4	Data sheet Table 1
DVDD_I/O	3.3 V $\pm$ 5%	40 mA	TB1 pin 2	Data sheet Table 1
VCC_USB	3.3 V	300 mA [‡]	TB1 pin 1	Estimated
GND	—	—	TB1 pin 3	—

[†] 185 mA device current plus up to 40 mA of DAC termination current, which is supply-referenced and sourced from AVDD.

[‡] Derived from the evaluation board schematic: Cypress CY7C68013 USB controller, two 74LVC541A buffers, eight 1 k Ω pull-downs, and a status LED. See §11.

2.2 Reference Clock Interface

The AD9959 reference clock input, with CLK_MODE_SEL low, is an AC-coupled buffer supporting single-ended or differential drive. The evaluation board implements the configuration shown in Figure 33 of the data sheet: a single-ended SMA input (J9) feeding a 1:1 transmission-line transformer (M/A-COM ETC1-1-13), a 50 Ω termination network, and 0.1 μ F coupling capacitors into REF_CLK (pin 23) and its complement (pin 22).

Table 2: Reference clock input specifications, AD9959 data sheet Table 1

Parameter	Min	Max	Note
REFCLK multiplier enabled	10 MHz	125 MHz	—
REFCLK multiplier bypassed	1 MHz	500 MHz	—
VCO output, gain bit high	255 MHz	500 MHz	Bit defaults low
VCO output, gain bit low	100 MHz	160 MHz	—
Input level	200 mV	1000 mV	Measured at each pin
Input bias	1.15 V typical		AC coupling mandatory
Input impedance	1500 Ω		—
Duty cycle, multiplier on	35%	65%	—
PLL multiplication factor	4	20	Integer

3 Architecture

3.1 Power Tree

All rails are linearly regulated. No switching converter appears anywhere in the design. Figure 1 shows the topology.

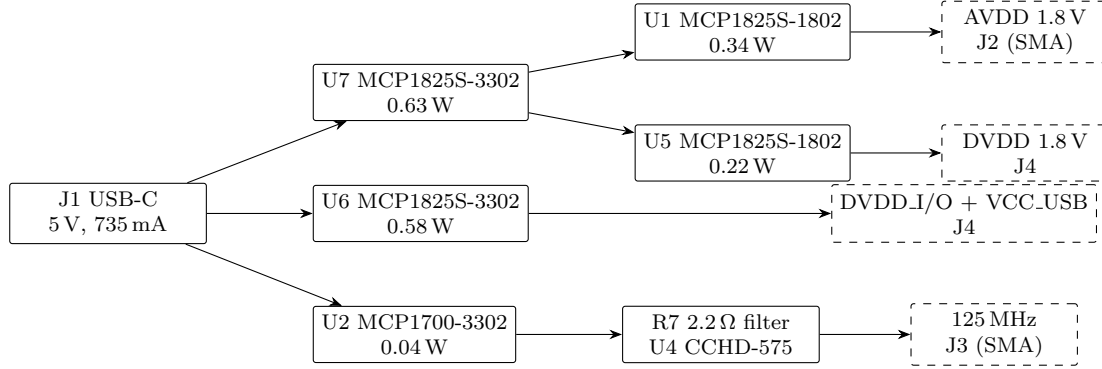


Figure 1: Power tree. Total dissipation 1.80 W; VBUS draw 735 mA. U7 and U6 are the thermal hot spots and are placed at the opposite end of the board from the oscillator.

The 3.3 V master rail (U7) feeds only the two 1.8 V regulators. The digital 3.3 V rail (U6) and the oscillator supply (U2) are independently regulated directly from VBUS. This separation keeps the USB controller’s switching current, which dominates the digital load, off both the analogue supply chain and the oscillator.

3.2 Reference Clock Chain

$$U4.3 \rightarrow R3 (100 \Omega) \rightarrow C2 (10 \text{ nF}) \rightarrow [R4 \mid R5 \mid R6] \rightarrow J3 \rightarrow \text{coax} \rightarrow \text{eval. J9}$$

R3 provides series termination and limits the oscillator output current. C2 is a DC block, required because the HCMOS output idles at $V_{dd}/2$ while the AD9959 self-biases its input at 1.15 V. R4/R5/R6 form a π attenuator footprint; R5 is populated as a 0 Ω jumper and the shunt positions are left unpopulated, permitting bench adjustment of the delivered level without board modification.

4 Design Rationale

4.1 Reference Frequency and Multiplier Selection

This is the governing design decision. The AD9959 data sheet specifies residual phase noise separately for each multiplier setting. At $f_{\text{OUT}} = 100.3 \text{ MHz}$ (Figures 16–18):

Table 3: AD9959 residual phase noise (SSB) versus REFCLK multiplier, $f_{\text{OUT}} = 100.3 \text{ MHz}$, $f_{\text{CLK}} = 500 \text{ MHz}$

Multiplier	1 kHz	10 kHz	100 kHz	1 MHz
Bypassed	−134	−144	−152	−154
5×	−120	−130	−135	−129
20×	−107	−119	−121	−119

All values dBc/Hz.

A naive design using a 25 MHz temperature-compensated oscillator at 20× multiplication yields a reference contribution of approximately −123 dBc/Hz at 1 kHz offset once translated to a 100 MHz output:

$$\mathcal{L}_{\text{out}} = \mathcal{L}_{\text{ref}} + 20 \log_{10}(M) + 20 \log_{10}\left(\frac{f_{\text{out}}}{f_{\text{CLK}}}\right) = -135 + 26.0 - 14.0 = -123 \text{ dBc/Hz.} \quad (1)$$

Against the 20× residual of −107 dBc/Hz, the device PLL dominates by 16 dB. *Improving the oscillator at 20× therefore yields no measurable benefit.*

Reducing the multiplication factor is the only effective lever. The minimum permitted factor is 4, requiring a 125 MHz reference—which is simultaneously the maximum permitted multiplier input frequency. This configuration recovers approximately 13 dB relative to 20×.

A secondary benefit follows from the higher system clock. For a 200 MHz output, the first Nyquist image moves from 130 MHz (at $f_{\text{CLK}} = 200 \text{ MHz}$) to 300 MHz, substantially relaxing the output reconstruction filter.

4.2 Oscillator Selection

With the multiplier fixed at 4×, the reference must be quieter than the device residual of approximately −122 dBc/Hz at 1 kHz. Candidates were evaluated against measured phase noise data referred to a 200 MHz DDS output (Table 4).

Table 4: Oscillator candidate comparison, 1 kHz offset, referred to a 200 MHz DDS output

Candidate	Technology	dBc/Hz	Disposition
AD9959 residual, 4×	—	−114	Reference floor
Crystek CCHD-575 @ 125 MHz	Quartz XO	−138	Selected
Abracon ABLNO @ 125 MHz	Quartz XO	−136	Second source
Taitien NA-100M-6900	OCCO	−151	Rejected: 2.1 W
SiTime SiT9501	MEMS	−113	Rejected: exceeds floor
SiTime SiT9365	MEMS	—	Rejected: no data published
Crystek CCHD-957	Quartz XO	—	Rejected: 10 to 50 MHz only

The MEMS candidates illustrate a common selection error: both publish excellent integrated RMS jitter (45 to 70 fs over 12 to 20 kHz–MHz) while exhibiting poor close-to-carrier performance, because the output derives from a fractional- N PLL locked to a low- Q resonator. Integrated jitter is a SerDes figure of merit and is not predictive of the 100 to 10 Hz–kHz region that governs AOM drive.

The selected part is the **Crystek CCHD-575-50-125.000**: 3.3 V HCMOS, ± 50 ppm, 0 to 70 °C, with manufacturer-measured phase noise of -141.9 dBc/Hz at 1 kHz offset at exactly 125 MHz. The ± 50 ppm/0 to 70 °C grade was selected over tighter options on availability grounds; phase noise is identical across grades, and ± 50 ppm corresponds to ± 10 kHz at a 200 MHz output, which is immaterial for the application. The restricted temperature grade is the reason the thermal separation described in §8.3 is required.

4.3 Oscillator Supply Filtering

The oscillator is powered by a dedicated regulator (U2) taken directly from VBUS, followed by an RC filter (R7, C5, C1). A ferrite bead was considered and rejected: simulation of a 600 Ω bead with the same capacitors predicts **30 dB of gain at 73 kHz**, because an LDO load draws near-constant current and therefore provides no damping of the resulting LC resonance. The RC alternative is unconditionally stable.

4.4 USB Hot-Plug Damping

Cable inductance (0.5 to 2 μ H) forms a series resonant circuit with the on-board ceramic capacitance (13.6 μ F effective after DC bias derating). Hot-plugging an energised source step-excites this tank. Without damping, simulation predicts a VBUS peak of 8.46 V against a regulator absolute maximum of 6.5 V.

A transient voltage suppressor alone is insufficient (7.31 V simulated), because it clamps only above 6.4 V and does not reduce the tank Q . The implemented solution is a damping branch: C6 (100 μ F polymer) in series with R8||R9 (0.235 Ω) to ground.

Critical: the damping is set by the resistors, not by capacitor ESR. C6 must be a low-ESR conductive polymer type. Substituting a wet aluminium electrolytic or tantalum with ESR near 1 Ω returns the circuit to a failing condition (6.9 V peak).

5 Circuit Description

5.1 Input Stage

J1 is a GCT USB4125-GF-A six-contact power-only USB Type-C receptacle. Both VBUS contacts (A9, B9) and both ground contacts (A12, B12) are paralleled, halving contact resistance. R1 and R2 present 5.1 k Ω R_d pull-downs on CC1 and CC2, identifying the board as a sink; both are required or a Type-C source will not enable VBUS. D1 (SMAJ5.0A) provides ESD and overvoltage protection. C6 and R8||R9 form the damping branch of §4.4.

5.2 Regulators

Table 5: Regulator assignment, loading and dissipation

Ref	Device	Load	Drop	Power	Tab net
U7	MCP1825S-3302E/DB	370 mA	1.7 V	0.63 W	GND
U6	MCP1825S-3302E/DB	340 mA	1.7 V	0.58 W	GND
U1	MCP1825S-1802E/DB	225 mA	1.5 V	0.34 W	GND
U5	MCP1825S-1802E/DB	145 mA	1.5 V	0.22 W	GND
U2	MCP1700T-3302E/MB	25 mA	1.7 V	0.04 W	VIN
Total		735 mA		1.80 W	

Assembly note: U2 is in a SOT-89 package whose tab is pad 2. On the MCP1700 this is **VIN**, not ground. It is the only regulator on the board whose tab is not at ground potential, and no ground pour may approach it. The four SOT-223 devices have ground tabs and share a common pour.

The MCP1825S was selected over the originally specified LD1117 for two reasons. First, it specifies 60 dB PSRR at 100 Hz, where the LD1117 data sheet specifies none; this matters because U7 feeds the AVDD regulator. Second, its tab is at ground potential, unifying the thermal pour across all four SOT-223 positions (the LD1117 tab is at VOUT).

5.3 Reference Clock Output Network

R3 (100 Ω) limits the CCHD-575 output driver to 20.2 mA against its ± 24 mA rating while delivering 5.33 dBm into 50 Ω . C2 (10 nF C0G) blocks the DC component. C0G dielectric is mandatory: X7R is piezoelectric and exhibits significant voltage coefficient, and this component lies in the 125 MHz signal path.

6 Verification

All simulations were performed in ngspice using component values and routed trace resistances extracted programmatically from the KiCad board file, ensuring the verified model corresponds to the fabricated artefact.

6.1 Reference Clock Delivery

Table 6: Reference clock levels versus oscillator output impedance, including 18.5 mm board trace and 300 mm coaxial jumper

R_{on}	Driver peak	Differential	Per pin	Level	Slew
14 Ω	20.9 mA	1.014 V	507 mV	5.69 dBm	0.50 V/ns
20 Ω	20.2 mA	0.978 V	489 mV	5.38 dBm	0.49 V/ns
30 Ω	19.0 mA	0.923 V	461 mV	4.88 dBm	0.46 V/ns
50 Ω	17.1 mA	0.830 V	415 mV	3.97 dBm	0.41 V/ns
96 Ω	13.8 mA	0.673 V	337 mV	2.17 dBm	0.34 V/ns

Crystek does not publish an output impedance. The range in Table 6 spans the values implied by the ± 24 mA drive specification (14 Ω) and by the 2 ns maximum rise time into the 15 pF reference load (96 Ω). Per-pin voltage remains within the 200 to 1000 mV specification across the entire range, and driver current never approaches the rating.

The square-wave drive is advantageous: the data sheet states that “for best phase noise performance, a clean, stable clock with a high slew is required.” An HCMOS output provides approximately 0.49 V/ns at the load, compared with roughly 0.35 V/ns for a 3 dBm sinusoid at the same frequency.

6.2 Hot-Plug Transient

Regulator absolute maximum is 6.5 V; the U7 dropout floor at 370 mA is 3.56 V. All corners pass with margin on both bounds.

The worst case is a *short*, low-resistance cable rather than a long one: low inductance combined with low series resistance produces the highest tank Q . Sensitivity analysis identifies source voltage (+0.60 V from 5.0 V to 5.5 V) and cable resistance (+0.42 V from 60 m Ω to 30 m Ω) as dominant; cable inductance contributes less than 0.15 V across its plausible range.

Table 7: VBUS peak across operating and tolerance corners, final component values, including routed trace resistance and the SMAJ5.0A clamp

Corner	Peak	Margin	Undershoot	$P/\text{resistor}$
Nominal, 5.0 V, 1 μH	4.94 V	+1.56 V	4.94 V	20.6 W
5.5 V, short low-loss cable	5.57 V	+0.93 V	4.87 V	36.9 W
Above, plus C6 -20% , R $+20\%$	5.87 V	+0.63 V	4.89 V	35.2 W
4.75 V, long cable, opposite tolerances	4.63 V	+1.87 V	4.63 V	9.3 W

R8 and R9 must both be populated. With a single resistor fitted, the peak rises to 6.77 V, exceeding the absolute maximum.

6.3 Oscillator Supply Filter

Table 8: R7–C5–C1 filter response, including measured via inductance

Frequency	3 vias (0.5 nH)	1 via (1.3 nH)
10 kHz	−2.3 dB	−2.3 dB
100 kHz	−18.5 dB	−18.5 dB
1 MHz	−40.1 dB	−40.1 dB
10 MHz	−33.1 dB	−33.0 dB
100 MHz	−25.4 dB	−17.9 dB
530 MHz	−7.8 dB	−4.6 dB

The -3 dB corner is 12.1 kHz with no gain peaking at any frequency. Supply voltage at the oscillator is 3.245 V, within the 3.0 to 3.6 V specification.

The three parallel ground vias at C1 contribute 7.5 dB of additional rejection at 100 MHz relative to a single via. This matters because the output driver’s switching current, with 600 ps edges, extends to approximately 530 MHz.

6.4 Rail Voltage Drop

Table 9: IR drop on routed rails, computed from extracted trace geometry

Net	Length	Width	R	Drop	% of rail
DIGITAL_3.3V	71.19 mm	0.6 mm	57.6 m Ω	19.6 mV	0.59%
+5V	47.72 mm	0.8 mm	29.0 m Ω	21.3 mV	0.43%
DVDD_1.8V	27.07 mm	0.6 mm	21.9 m Ω	3.2 mV	0.18%
3.3V_Master	22.51 mm	0.6 mm	18.2 m Ω	6.7 mV	0.20%
AVDD_1.8V	10.20 mm	0.6 mm	8.3 m Ω	1.9 mV	0.10%
OSC_3.3V	3.75 mm	0.4 mm	4.6 m Ω	0.1 mV	0.00%

7 External Interconnection

7.1 AVDD Coaxial Jumper

This is the most constrained external connection. The AD9959 specifies AVDD as $1.8\text{ V} \pm 5\%$, giving a floor of 1.710 V. The MCP1825S fixed-output tolerance is $\pm 2.5\%$ maximum, placing the regulator output as low as 1.755 V. Only 45 mV remains for interconnect loss at 225 mA, corresponding to a maximum total loop resistance of 192 m Ω .

Table 10: Worst-case AVDD at the evaluation board versus cable type and length

Cable	Loop Ω/m	150 mm	300 mm	600 mm
RG-178	1.150	1.714 V	1.676 V	—
RG-316	0.370	1.741 V	1.728 V	1.703 V
RG-400	0.041	1.752 V	1.750 V	1.748 V

RG-316 and RG-178 employ copper-clad steel centre conductors with approximately ten times the DC resistance of solid copper. **RG-400 is specified** for this connection; if unavailable, RG-316 is acceptable at 300 mm or shorter. RG-178 must not be used.

7.2 Supply Harness

J4 mates to evaluation board terminal block TB1. Wire must be AWG 24 or heavier and no longer than 300 mm. The ground conductor carries the sum of all three rails (485 mA); at AWG 24 over 200 mm this produces 5 mV of common-mode offset, rising to 31 mV at AWG 28 over 300 mm.

Critical: DVDD has an absolute maximum of 2.0 V and is adjacent to two 3.3 V rails on the same connector. No protection on either board prevents a transposed conductor from destroying the AD9959. Label every conductor and verify continuity with a multimeter before first connection.

8 PCB Implementation

8.1 Stack-up

Table 11: Four-layer stack-up, 1.58 mm nominal

Layer	Thickness	Material	Function
F.Cu	35 μm	Cu 1 oz	Components, RF, local power
—	0.1 mm	FR-4, $\epsilon_r = 4.5$	Prepreg
In1.Cu	35 μm	Cu 1 oz	Unbroken ground plane
—	1.24 mm	FR-4, $\epsilon_r = 4.5$	Core
In2.Cu	35 μm	Cu 1 oz	+5V and 3.3V_Master pours
—	0.1 mm	FR-4, $\epsilon_r = 4.5$	Prepreg
B.Cu	35 μm	Cu 1 oz	Ground pour, thermal

Board outline is 60 mm square with four M3 mounting holes. Surface finish is ENIG, specified by the CCHD-575 data sheet (“PAD FINISH: Immersion Gold (ENIG); 5 micro inches maximum”) and advantageous for the planar thermal tabs.

The 0.1 mm prepreg between F.Cu and In1.Cu provides tight coupling: approximately half the return-loop inductance of a 0.2 mm build, plus roughly 600 pF of distributed plane capacitance. In1.Cu carries no routing and no zone cut-outs.

8.2 Controlled Impedance

Controlled impedance was not specified for fabrication. At 125 MHz the critical length for a 600 ps edge is 49 mm; the reference clock trace is 18.5 mm, so transmission-line behaviour is not engaged. Simulation across a 2.4:1 impedance range (21 to 50 Ω) showed a 0.14 dB spread in delivered level and no ringing. The 0.25 mm trace width used corresponds to approximately 38 Ω on this stack-up.

8.3 Thermal Design

Total dissipation is 1.80 W. On a four-layer 60 mm board with 1 oz inner copper, the estimated rise is 29 °C, placing the board at approximately 54 °C in a 25 °C ambient. The CCHD-575 is a 0 to 70 °C grade part, so this margin is adequate but is the reason U4 is placed 38 to 43 mm from U6 and U7.

Thermal vias are placed *outside* the regulator tab pads rather than within them, avoiding solder wicking during reflow. Radial spreading analysis indicates this offset costs 2 to 4 °C, approximately 5–7% of the total thermal path.

9 Assembly

9.1 Reflow

SAC305 (Sn96.5/Ag3.0/Cu0.5), no-clean ROL0 flux, Type 4 or finer powder. The profile is bounded by the CCHD-575 data sheet:

Table 12: Reflow profile

Ramp-up	$\leq 3\text{ °C/s}$ (Crystek limit)
Soak	150 to 200 °C, 60 to 120 s
Above liquidus	45 to 75 s above 217 °C
Peak	240 °C (Crystek limit)
Ramp-down	$\leq 6\text{ °C/s}$ (Crystek limit)

All components clear this peak: the USB4125 is rated to 255 °C and the OS-CON polymer capacitor to 260 °C. Crystek specifies oscillator frequency “at 25 °C, 1 hour after reflow”; frequency measurements taken immediately after reflow will not reflect the settled value.

9.2 Critical Placement

1. C1 within 2 mm of U4 pin 4, with dedicated ground vias. Measured: 1.85 mm, three vias within 1.83 mm.
2. No ground pour beneath U2’s tab (at VIN).
3. Both R8 and R9 populated.
4. R5 populated; it is the only signal path from the oscillator to J3.

10 Bring-Up Procedure

1. Power the board with no load attached. Verify all five rails.
2. Measure VBUS at J1 under full load with the intended charger and cable.
3. Observe J3 unterminated, then into 50 Ω. Expect 5.3 dBm and 0.98 V peak-to-peak into 50 Ω. A high-impedance probe will indicate approximately 3.3 V peak-to-peak; this is expected and not a fault.
4. Verify the evaluation board reference clock termination: measure from the balun secondary to ground. Approximately 18.75 Ω indicates all three resistors are fitted (a 2:1 mismatch); remove R10.
5. Read the marking on the evaluation board’s Cypress device (see §11).
6. Measure VCC_USB current at TB1 pin 1 with the control software active.

7. Connect the harness. **Verify all four conductors with a multimeter before applying power.**
8. Set jumper W11 to the REF CLK position.
9. Configure the AD9959: multiplier = 4, **VCO gain control bit set high**. The bit defaults low, covering only 100 to 160 MHz; a 500 MHz system clock requires the 255 to 500 MHz range. Use the lowest charge-pump current that locks reliably, as higher settings increase phase noise.

Power the board from a dedicated USB-C wall adapter, not from the host computer. The 735 mA draw exceeds the 500 mA available from a legacy USB-A port, and a separate adapter breaks the ground loop formed through the evaluation board's USB connection to the host.

11 Open Items and Assumptions

The following quantities are estimated rather than specified. The design passes across the plausible range of each, but they should be confirmed empirically.

1. **VCC_USB current.** Estimated 300 mA worst case from the evaluation board schematic. If the Cypress device is marked CY7C68013A (FX2LP) rather than CY7C68013 (FX2), the load is approximately 110 mA and U6 dissipation falls from 0.58 W to 0.25 W. Analog Devices publishes no current figure for this rail.
2. **Oscillator output impedance.** Not published by Crystek. The verification in Table 6 spans 14 to 96 Ω ; all cases pass.
3. **Cable inductance.** Modelled at 0.5 to 2 μH . The damping design passes across this range; the parameter sets the hot-plug peak.
4. **AD9959 residual phase noise at 200 MHz.** Extrapolated by $20 \log_{10}$ from published 100.3 MHz data. The scaling reproduced the measured $5\times/20\times$ difference to within 1 dB.
5. **Evaluation board termination network.** The populated configuration of R10/R30/R31 has not been confirmed on the physical board.

A Bill of Materials

Ref	Value	Manufacturer part	Note
C1, C2	10 nF	Murata GRM1555C1H103JA01D	C0G mandatory , 50 V
C3, C9–C17	4.7 μF	Murata GRM188R61E475KE11D	X5R 25 V
C4	1 μF	TCC0402X7R105K160AT	X7R 16 V
C5	10 μF	Murata GRM21BZ71E106KE15L	X7R 25 V
C6	100 μF	Panasonic 16SVPC100M	Polymer only
D1	—	Littelfuse SMAJ5.0A	TVS, cathode to +5V
D2	3.6 V	onsemi MMSZ5227BT1G	3.6 V, not 3.3 V
J1	—	GCT USB4125-GF-A	USB-C, power only
J2, J3	—	Adam Tech RF2-04A-T-00-50-G	SMA jack
J4	—	Samtec TSW-104-07-G-S	1 $\times$ 4, 2.54 mm
R1, R2	5.1 k Ω	Yageo RC0402FR-075K1L	USB-C R_d
R3	100 Ω	Panasonic ERA-3AEB101V	Thin film
R4, R6	DNP	—	π pad shunts
R5	0 Ω	Yageo RC0603JR-070RL	Must be populated

Ref	Value	Manufacturer part	Note
R7	2.2 Ω	Yageo RC0603FR-072R2L	Oscillator filter
R8, R9	0.47 Ω	Vishay RCWE1206R470FKEA	Pulse rated; both required
U1, U5	—	Microchip MCP1825S-1802E/DB	1.8 V LDO
U2	—	Microchip MCP1700T-3302E/MB	Tab = VIN
U4	—	Crystek CCHD-575-50-125.000	125 MHz XO
U6, U7	—	Microchip MCP1825S-3302E/DB	3.3 V LDO

B Derivation of the Damping Requirement

The cable inductance L_c and on-board ceramic capacitance C_c form a series resonant circuit of characteristic impedance

$$Z_0 = \sqrt{\frac{L_c}{C_c}} = \sqrt{\frac{1 \mu\text{H}}{13.6 \mu\text{F}}} = 0.271 \Omega. \quad (2)$$

Critical damping requires a total series resistance of $2Z_0 = 0.542 \Omega$. Cable resistance contributes only 30 to 150 m Ω , leaving the tank substantially underdamped and producing the 8.46 V peak observed in simulation.

The damping branch cannot be placed in series with the supply, as this would introduce a DC drop at the full 735 mA load. It is instead AC-coupled: C6 presents a low impedance at the resonant frequency (0.026 Ω at 61 kHz for 100 μF) while blocking DC, allowing R8||R9 to damp the tank without steady-state dissipation.

The resistor value is bounded on both sides. Too low, and insufficient energy is dissipated per cycle. Too high, and C6 is effectively decoupled from the tank. Simulation identifies a usable window of 0.10 to 0.60 Ω with an optimum near 0.235 Ω .

Peak resistor dissipation follows from the branch current:

$$P_{\text{peak}} = \left(\frac{I_{\text{peak}}}{2} \right)^2 R_{\text{each}} = \left(\frac{17.7 \text{ A}}{2} \right)^2 \times 0.47 \Omega \approx 36.9 \text{ W}, \quad (3)$$

with energy $E = P_{\text{peak}}\tau/2 \approx 655 \mu\text{J}$ for $\tau = RC_6 = 23.5 \mu\text{s}$. The equivalent rectangular pulse duration is $E/P_{\text{peak}} = 13.4 \mu\text{s}$. Resistor selection therefore requires a published single-pulse power curve clearing 37 W at 13 μs ; steady-state power rating is irrelevant because C6 blocks DC.